Anuj Pratap Singh Yadav

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SUMMARY

Staff engineer, 11 years of C++ systems at Synopsys, Microsoft and Adobe. I own coverage analytics in a verification product, lead cross-team architecture in a large C++ codebase, and build LLVM static analysis and AI agents.

SKILLS

Languages: C++, Python, Tcl, C#, Java, TypeScript

AI-assisted engineering: Cursor agents, skills, rules and subagents; scheduled Jira automation

Architecture: dependency inversion, IWYU, LLVM IR passes, master/worker orchestration, XML (pugixml)

Distributed systems (coursework, prototypes): microservices, Kafka, Redis, caching, sharding, consistent hashing

Tools: GDB, Linux, LSF, Perforce, Git, Win32

EXPERIENCE

R&D Staff Engineer | Synopsys *Noida, India | May 2023 - Present*

VC Formal, a verification product for chip designs; my app checks a C++ model against its hardware (RTL) implementation

- **Build-dependency decoupling in a large C++ codebase (multi-org)**: led VC Formal's part. The shared C++ layer under ~15 apps and tools included engine headers, so one engine edit rebuilt every app. Cut wrong-way dependencies ~73% with include-what-you-use, hide-what-you-use (forward declarations), dependency inversion through startup-injected callbacks, and code relocation, in small changes reviewed by the owning teams; in production. Org-wide tech talk on the design.
- **AI agents for the team's workflow**: built Cursor agents teammates use: a daily Jira check that flags tickets with wrong validation fields and notifies owners, and an agent that reproduces a Jira issue, root-causes it and drafts a fix as a code review for an engineer to check.
- **Coverage analytics for C++ code**: own it end to end. It shows which parts of a C++ model the automated proofs actually exercised, so C++ can be signed off like hardware code. Added merging across many runs and export to the standard coverage database, so C++ and hardware coverage are reviewed together in one viewer; handle customer issues and guided its integration into a new sign-off app.
- **LLVM static analysis (customer-requested R&D)**: building program slicing that reports the exact C++ source lines a checked property depends on. Built on the open-source dg LLVM slicer: wrote LLVM IR passes for C++ constructs it missed and a signal-based slicing criterion that cut whole-file slices down to the lines feeding each signal. Design reviewed weekly with a Distinguished Engineer.
- **Distributed job orchestration: license management**: built token licensing in the C++ engine and moved runtime license checkout from workers to the master of a master/worker job runner, so each key serves more workers; fixed production checkout failures across license servers.
- **Coverage analytics for hardware designs**: own property density and RTL coverage, which report which signals, down to the bit, and which lines the checks actually exercise. Rebuilt the density analysis on a shared design model so both apps give the same answer, adding bit-level detail; aligned coverage semantics across both. In production.
- **Customer escalations**: own them end to end in my areas: root cause, regression test, fix shipped in service packs.
- **Session persistence**: closed silent state-loss gaps (Tcl procedures, engine state as XML), with round-trip tests.

Software Engineer II | Microsoft *Noida, India | May 2021 - May 2023*

- Owned Recording Studio features in PowerPoint Win32 (C++): camera modes, live background blur, video export and Cameo, a camera object placed on the slide. Led v1 test automation and wired Recording and Export telemetry.
- Designed catalog filtering plus an allowlist (C#) so new commerce SKUs no longer broke OneDrive's upgrade path.
- Built the Stream 2.0 video playback component in Excel Win32 (C++, TypeScript) for SharePoint and OneDrive video.

Software Development Engineer 2 | Adobe *Noida, India | Aug 2015 - Apr 2021*

- Moved Photoshop and Premiere Elements from RIBS to Hyperdrive builds with parallel bundles, cutting install time ~50%.
- Designed lazy loading for large files in the AEM web editor, cutting load time ~60% on files customers reported as freezing.

EDUCATION AND AWARDS

Spot Awards, Synopsys (latest Sep 2026, org all-hands) | Spot Award, Adobe (RIBS to Hyperdrive migration)

B.Tech, Computer Science and Engineering - NIT Bhopal (MANIT) | 2011 - 2015 | GPA 8.27/10